

Arithmetic operators

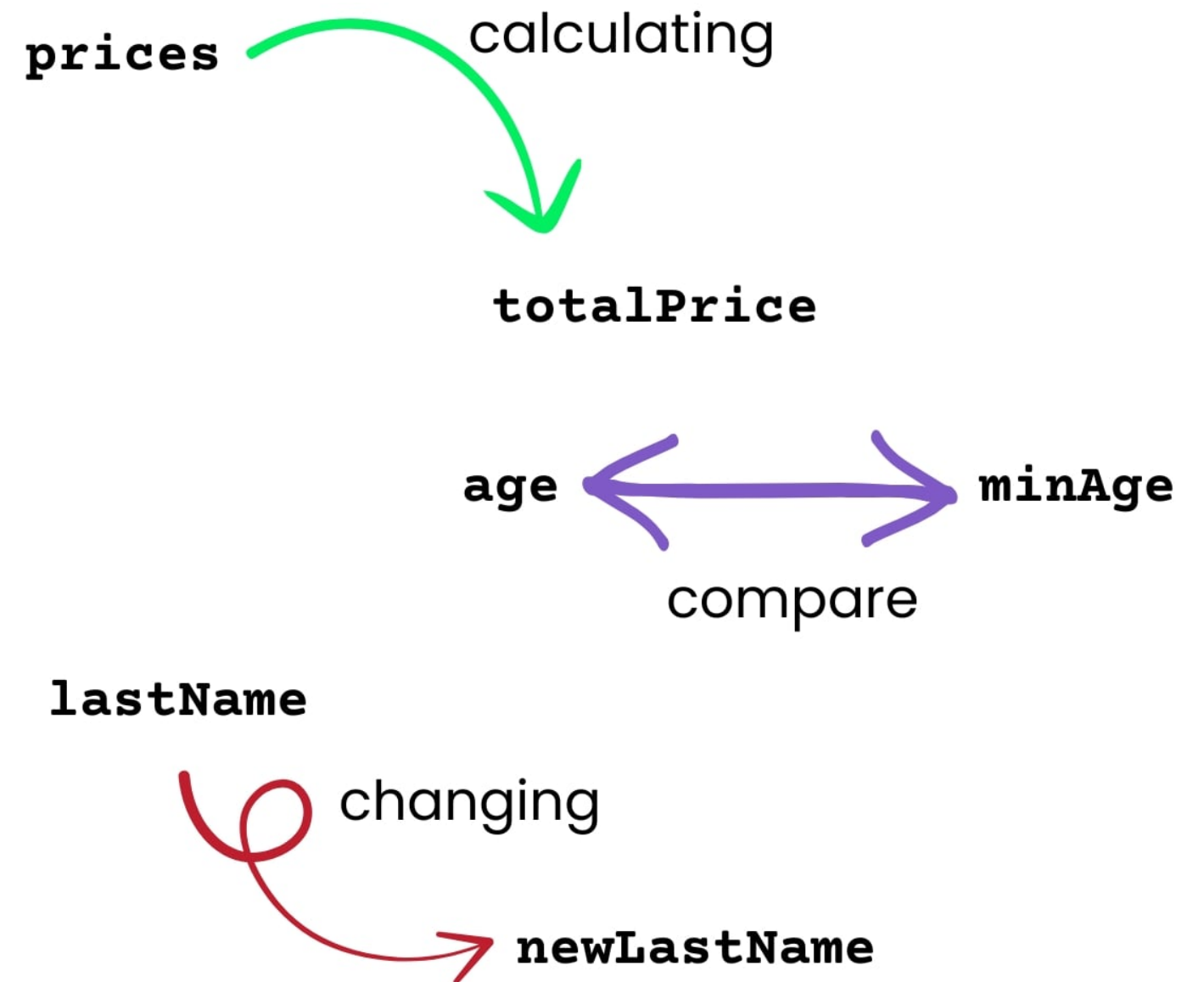
INTRODUCTION TO JAVA



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Operators

- Working with, modifying variables
 - Performing calculations
 - Changing values
 - Comparing values
- Operators: built-in symbols for various tasks
- Arithmetic operators



Addition using +

```
public class Main {  
    public static void main(String[] args) {  
        // Assign calculation to variable a  
        int a = 5 + 6;  
        // Print the value of a  
        System.out.println(a);  
    }  
}
```

Addition using +

```
// Adding doubles
double x = 5.5;
double y = 4;

double xPlusY = x + y; // Value is 9.5

// String concatenation
String message = "Hello";
String messageForJim = message + " Jim"; // Value is "Hello Jim"
```

Increment using ++

- Use ++ to raise value by 1

```
int age = 30;
```

```
// Using increment to add 1
```

```
age++; // Value is now 31
```

Subtraction using -

```
// Subtract
```

```
double productPrice = 99.9;
```

```
productPrice = productPrice - 10; // Value is now 89.9
```

```
// Decrement value
```

```
int productLaunch = 99;
```

```
productLaunch--; // Values is now 98
```

Multiply using *

```
int price = 100;
int quantity = 1;

// Multiply two integers
int orderRevenue = newPrice * quantity; // Value is 100, int

double newPrice = 94.9;
int quantity = 1;

// Multiply with double
double newOrderRevenue = newPrice * quantity; // Value is 94.9, double
```


Division using /

```
class IntDivision {  
    public static void main (String[] args){  
        int numOrders = 100;  
        int days = 30;  
        // Output is 3,  
        // because both numbers are integers  
        System.out.println(numOrders / days);  
    }  
}
```

3

Division using /

```
class DoubleDivision {  
    public static void main (String[] args){  
        int numOrders = 100;  
        int days = 30;  
        // Either assing to double  
        double avgOrders = numOrders / days;  
        System.out.println(avgOrders);  
        // Or use a double in the division  
        System.out.println(numOrders / 30.0);  
    }  
}
```

3.3333333333333335

Recap

Operation	Operator	Example
Addition	+	5 + 1.5
Subtraction	-	3.6 - 2
Multiplication	*	0.3 * 2
Division	/	59 / 4

- Division of two `int` s returns an `int` , unless assigned to `double`
- Division with at least one `double` returns a `double`

Let's practice!

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Comparison operators

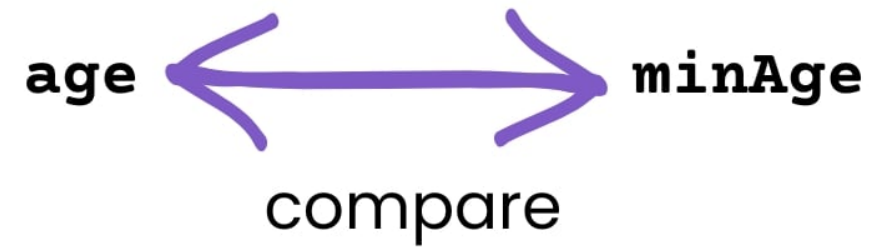
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Comparisons

- Always evaluate to `true` or `false` (result is a `boolean`)



Greater than >, less than <

```
class GreaterThan {  
    public static void main(String[] args){  
        System.out.println(5 < 6); // Will print true, 5 is less than 6  
    }  
}
```

true

Greater than >, less than <

```
// Check if 5 is less than 6
```

```
boolean x = 5 < 6; // Value is true, because 5 < 6
```

```
// Check if 5 is greater than 6
```

```
boolean y = 5 > 6; // Value is false, 5 < 6
```


Greater or equal to >=

```
int minSpend = 25;  
int total = 25;  
  
// Check if total is greater or equal to minSpend  
boolean freeDelivery = total >= minSpend; // Value is true, total is equal to minSpend
```


Less or equal to <=

```
int minSpend = 25;  
int total = 23;  
  
// Check if total is less or equal to minSpend  
  
boolean paidDelivery = total <= minSpend; // Value is true, total is less than minSpend
```

Equal ==

- We need to use `==` , `=` is for assignment

```
int userAccountNumber = 567346;  
int submittedAccountNumber = 456777;  
  
// Check if userAccountNumber is equal to submittedAccountNumber  
boolean isUserAccountNumber = userAccountNumber == submittedAccountNumber;  
// Value is false
```

Not equal !=

- Use !=

```
int userHealth = 235;

// Check whether userHealth is not zero
boolean alive = userHealth != 0; // Value is true, userHealth is not zero
```

Comparison summary

Operator	Name	Example	Result of example
>	Greater than	6 > 6	false
<	Less than	5 < 6	true
>=	Greater or equal to	5 >= 6	false
<=	Less or equal to	5 <= 6	true
==	Equal to	5 == 5	true
!=	Not equal to	5 != 5	false

- Output of all operations is a boolean

Let's practice!

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Wrap-up

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Recap

What we learned:

- What is Java and why it's an important programming language
- How to create Java applications using `class`
- Java primitives and primitive operators
- Declaring variables
- What, when and why around Strings and Arrays

What's next?

- Internet Sources
 - **Baeldung**: website dedicated to Java topics and assistance
 - **Oracle's Java site**: the stewards of Java technology (Oracle) web site
 - **dev.java**: Java developer news, events, and other information
 - **Open JDK**: community dedicated to a free and open-source implementation of Java
- Books
 - *Head First Java* by Sierra et al
 - *Learn Java In One Day* by Jamie Chan
 - *Java for Beginners* by Brady Ellison
- Additional online course from **DataCamp**
 - **Intermediate Java**

Congratulations!

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